

The Resolved-Face Crown of the Dual-Rail Carrier: Self-Dual Completeness, the Exact Count, and One-Constant Completion

Won Chul Yang

Independent Researcher, Seoul, Republic of Korea

wcy0969@gmail.com

Public edition v1.0 — 2026

Companion note to “The Price of NOT on D4” (Paper II, DOI: 10.5281/zenodo.21800033): it proves in full the two crown facts that Paper II’s Appendix A.4–A.5 uses as boundary context and attributes to a companion line. Every headline result is machine-verified in dependency-minimal Lean 4 (core only; no mathlib; no `native_decide`; no `sorry`; kernel axiom profile at most `[propext, Quot.sound]`).

Abstract

On the four-state dual-rail carrier D4 with closed core $G = \text{Clo}\{\wedge_k, \vee_k, P\}$, consider the resolved face $R = \{\text{FAL}, \text{TRU}\}$ and the outer fragment $G|_R$ of R -valued restrictions of closed-core terms on R^n . We prove the crown theorem: $G|_R^{(n)}$ is exactly the set of P -equivariant maps $R^n \rightarrow R$ — equivalently, under the Boolean reading of the face, exactly the self-dual Boolean functions of arity n . The count is therefore $2^{2^{n-1}}$: the diagonal P -action on R^n is free, and an equivariant map freely chooses one output per orbit. The self-dual reading corrects a natural miscount: 2^n counts per-coordinate routing data, not functions — odd-arity majority is self-dual and lies in the crown, while AND, OR, and both constants do not. We then prove one-constant completion: adjoining a single resolved constant to the crown realizes every map $R^n \rightarrow R$, by an explicit self-dualization with one guard variable. Finally we record why the crown is a genuinely dual-rail (single-orbit) phenomenon: for resolved faces with two or more P -orbits, P -equivariance strictly over-approximates the outer fragment, witnessed by an explicit unary counterexample.

Keywords: clone theory; self-dual Boolean functions; dual-rail carriers; equivariant maps; functional completeness; Lean 4.

MSC 2020: 03B50, 06D30, 08A40, 94C10.

1 Introduction

The companion paper [5] studies the four-state dual-rail carrier $\text{D4} = \{\text{UNK}, \text{FAL}, \text{TRU}, \text{CON}\}$ with closed core $G = \text{Clo}\{\wedge_k, \vee_k, P\}$ and proves the exact negation-complexity law $\nu(f) = \text{dec}(f)$ on the resolved face $R = \{\text{FAL}, \text{TRU}\}$ after the confinement-breaking operation σ_s is adjoined. Before σ_s enters, the closed core already induces an *outer fragment* on the resolved face:

$$G|_R^{(n)} = \{t|_{R^n} : t \text{ an } n\text{-ary term of } G, t(R^n) \subseteq R\},$$

the family of R -valued restrictions of closed-core terms. Paper II’s Appendix A.4–A.5 uses two facts about this fragment as boundary context — that it consists exactly of the P -equivariant maps and has cardinality $2^{2^{n-1}}$, and that one adjoined resolved constant completes it to all maps $R^n \rightarrow R$ — attributing both to a companion line. This note is that companion: it states and proves both facts in full, together with the structural reading that makes them stable, and it machine-verifies the results in Lean 4 on top of the published mechanization of [5].

Three points deserve emphasis.

The self-dual reading. Under the Boolean encoding $\text{TRU} = 1$, $\text{FAL} = 0$ of the face, P acts as negation, and P -equivariance of $h : R^n \rightarrow R$ becomes $h(\neg x) = \neg h(x)$: the crown is exactly the clone-theoretic class of *self-dual* Boolean functions [3, 2]. This immediately calibrates what the crown does and does not exclude: AND, OR, and the constants are not self-dual and are excluded; odd-arity majority *is* self-dual and is included, realized inside the carrier as orbit-dependent routing (Proposition 4.1 gives the explicit ternary-majority term). In particular, the correct count of the crown is the count of self-dual functions, $2^{2^{n-1}}$ — not 2^n , which counts per-coordinate routing data rather than functions; at $n = 3$ the crown has 16 elements, including majority, against $8 = 2^3$ (Remark 4.2).

The two-layer proof. Necessity is inherited: closed-core terms are P -equivariant on all of $D4^n$ by the qualitative soundness theorem of [5], so their R -valued restrictions are P -equivariant on the face. Sufficiency is an explicit construction: one orbit-literal minterm per TRU -point of the target map, joined; on the face, the escape values UNK of the mismatched minterms are absorbed by the joins, and the single matched minterm (or the matched anti-minterm) delivers the resolved output. No step uses σ_s .

The boundary. The crown is a single-orbit phenomenon. On resolved faces with two or more P -orbits, a resolved value carries both a within-pair bit and an orbit label; P -equivariance constrains only the former, while closed-core terms are confined to orbits. Section 6 records the minimal unary counterexample. This is why the crown theorem is stated for the dual-rail carrier and claimed for nothing larger.

2 Preliminaries

We use the setting of [5, §2]. $D4$ is identified with $\{0, 1\}^2$ via $\text{UNK} = (0, 0)$, $\text{FAL} = (0, 1)$, $\text{TRU} = (1, 0)$, $\text{CON} = (1, 1)$; $\wedge_k$ and $\vee_k$ are componentwise meet and join in the knowledge order; $P(a, b) = (b, a)$. The resolved face is $R = \{\text{FAL}, \text{TRU}\}$, on which P restricts to the transposition. Under the Boolean encoding $\text{TRU} = 1$, $\text{FAL} = 0$, a map $h : R^n \rightarrow R$ is P -equivariant exactly when it is self-dual: $h(\neg x) = \neg h(x)$.

We use one published fact from [5], machine-verified there:

- (E) (Theorem 2.1 of [5]) every term operation of G is monotone for the knowledge order, preserves both endpoints, and is P -equivariant on $D4^n$.

We also use the elementary evaluation table of the internal operations on resolved and endpoint values (immediate from the componentwise definitions, and machine-checked in the present artifact):

$$\text{TRU} \wedge_k \text{TRU} = \text{TRU}, \quad \text{FAL} \wedge_k \text{FAL} = \text{FAL}, \quad \text{TRU} \wedge_k \text{FAL} = \text{UNK},$$

$\text{UNK} \wedge_k v = \text{UNK}$, $\text{TRU} \vee_k \text{UNK} = \text{TRU}$, $\text{FAL} \vee_k \text{UNK} = \text{FAL}$, $\text{UNK} \vee_k \text{UNK} = \text{UNK}$,
together with P -equivariance of $\wedge_k, \vee_k$ and $P(\text{FAL}) = \text{TRU}$.

3 The Crown Theorem

Theorem 3.1 (crown; self-dual completeness of the outer fragment). *For every arity $n \geq 1$,*

$$G|_R^{(n)} = \{h : R^n \rightarrow R \mid h \text{ is } P\text{-equivariant}\} = \{\text{self-dual Boolean functions of arity } n\}.$$

Proof. The second equality is the Boolean reading of Section 2. We prove the first.

($\subseteq$) Let t be an n -ary closed-core term with $t(R^n) \subseteq R$. By (E), t is P -equivariant on $\mathbf{D4}^n$; restricting to R^n (which is P -invariant) gives a P -equivariant map $R^n \rightarrow R$.

($\supseteq$) Let $h : R^n \rightarrow R$ be P -equivariant. First, h attains the value TRU : P -equivariance gives $h(P\vec{x}) = Ph(\vec{x}) \neq h(\vec{x})$, so h is nonconstant, and a map into R that is nonconstant attains both values. Let $A = \{\vec{y} \in R^n : h(\vec{y}) = \text{TRU}\}$; A is nonempty, and $P(A)$ is exactly the set where $h = \text{FAL}$ (equivariance), so A contains no P -pair: $\vec{y} \in A \Rightarrow P\vec{y} \notin A$.

For each $\vec{y} \in A$ define the *orbit-literal minterm*

$$M_{\vec{y}}(\vec{x}) = \bigwedge_k \{\ell_1(x_1), \dots, \ell_n(x_n)\}, \quad \ell_i(x_i) = \begin{cases} x_i, & y_i = \text{TRU}, \\ Px_i, & y_i = \text{FAL}, \end{cases}$$

a closed-core term (only $\wedge_k$ and P occur). Evaluate $M_{\vec{y}}$ on a face point $\vec{z} \in R^n$. Each literal value $\ell_i(z_i)$ is resolved, and equals TRU exactly when $z_i = y_i$. On resolved values, $\text{TRU} \wedge_k \text{TRU} = \text{TRU}$, $\text{FAL} \wedge_k \text{FAL} = \text{FAL}$, $\text{TRU} \wedge_k \text{FAL} = \text{UNK}$, and UNK absorbs. Hence:

- if $\vec{z} = \vec{y}$, every literal is TRU and $M_{\vec{y}}(\vec{z}) = \text{TRU}$;
- if $\vec{z} = P\vec{y}$, every literal is FAL and $M_{\vec{y}}(\vec{z}) = \text{FAL}$;
- otherwise the literals are mixed and $M_{\vec{y}}(\vec{z}) = \text{UNK}$.

Now set $t = \bigvee_k \{M_{\vec{y}} : \vec{y} \in A\}$ (a join over the nonempty family A ; again a closed-core term). Evaluate t on $\vec{z} \in R^n$:

- if $h(\vec{z}) = \text{TRU}$, then $\vec{z} \in A$, so the minterm $M_{\vec{z}}$ contributes TRU ; every other $\vec{y} \in A$ has $\vec{y} \neq \vec{z}$ and also $P\vec{y} \neq \vec{z}$ (else $\vec{z} = P\vec{y}$ with $\vec{y} \in A$, so $h(\vec{z}) = h(P\vec{y}) = Ph(\vec{y}) = \text{FAL}$, contradiction), so every other minterm contributes UNK . Since $\text{TRU} \vee_k \text{UNK} = \text{TRU}$ and $\text{UNK} \vee_k \text{UNK} = \text{UNK}$, the join is $\text{TRU} = h(\vec{z})$.
- if $h(\vec{z}) = \text{FAL}$, then $P\vec{z} \in A$, so the minterm $M_{P\vec{z}}$ evaluates at $\vec{z} = P(P\vec{z})$ to FAL ; every other $\vec{y} \in A$ has $\vec{y} \neq \vec{z}$ (as $h(\vec{y}) = \text{TRU}$) and $\vec{y} \neq P\vec{z}$, so every other minterm contributes UNK . Since $\text{FAL} \vee_k \text{UNK} = \text{FAL}$ and no minterm contributes TRU (that would require $\vec{z} \in A$), the join is $\text{FAL} = h(\vec{z})$.

In both cases $t(\vec{z}) = h(\vec{z}) \in R$. Hence $t|_{R^n} = h$ with $t(R^n) \subseteq R$, so $h \in G|_R^{(n)}$. $\square$

Remark 3.2. The construction realizes h with a term whose only generators are $\wedge_k, \vee_k, P$ — one orbit-literal minterm per TRU -point of h . An alternative route extends the face labeling monotonically through the ambient rail cube (the face is a one-hot antichain in $\{0, 1\}^{2n}$) and lifts the resulting positive formula; the direct construction above is the one mechanized.

Theorem 3.3 (the exact count). *For every $n \geq 1$, $|G|_R^{(n)}| = 2^{2^{n-1}}$.*

Proof. By Theorem 3.1 it suffices to count P -equivariant maps $h : R^n \rightarrow R$. The diagonal P -action on R^n is free: P has no fixed point on R , so $P\vec{x} = \vec{x}$ is impossible already in the first coordinate. Hence R^n (of size 2^n) splits into 2^{n-1} orbits of size two; each orbit has a unique representative with first coordinate TRU. A P -equivariant h is freely determined by its values on the 2^{n-1} representatives — the value on the partner point is forced to be the P -image — and every such assignment defines an equivariant map. This is a bijection between $G|_R^{(n)}$ and the maps from the representatives to R , of cardinality $2^{2^{n-1}}$. $\square$

For $n = 1, \dots, 5$ the counts are 2, 4, 16, 256, 65,536.

4 What the Crown Contains — and the Corrected Count

Proposition 4.1 (calibration). *Under the Boolean reading:*

1. *AND, OR, and both constants do not lie in $G|_R$ (none is self-dual; constants are moreover not P -equivariant).*
2. *Odd-arity majority lies in $G|_R$. Explicitly, for $n = 3$, with the agreement detectors*

$$D_{\text{same}}(x, y) = (x \wedge_k y) \vee_k (Px \wedge_k Py), \quad D_{\text{opp}}(x, y) = D_{\text{same}}(x, Py),$$

which are E -valued on R^2 ($E = \{\text{UNK}, \text{CON}\}$) with $D_{\text{same}} = \text{CON}$ iff $x = y$ and $D_{\text{opp}} = \text{CON}$ iff $x = Py$, the term

$$\text{MAJ}_3(x_1, x_2, x_3) = (x_1 \wedge_k (D_{\text{same}}(x_1, x_2) \vee_k D_{\text{same}}(x_1, x_3))) \vee_k (Px_1 \wedge_k (D_{\text{opp}}(x_1, x_2) \wedge_k D_{\text{opp}}(x_1, x_3)))$$

is a closed-core term whose restriction to R^3 is ternary majority.

Proof. (1) Self-duality $f(\neg x) = \neg f(x)$ fails for AND at $x = (1, 0)$ (it would force $\text{AND}(0, 1) = \neg \text{AND}(1, 0)$, i.e. $0 = \neg 0$), and dually for OR; constants c satisfy $Pc \neq c$ on R , so P -equivariance fails at once. (2) Majority of odd arity satisfies $\text{MAJ}(\neg x_1, \dots, \neg x_m) = \neg \text{MAJ}(x_1, \dots, x_m)$ (complementing all inputs complements the strict majority), so it is self-dual and lies in the crown by Theorem 3.1. For the explicit term: on resolved inputs the detectors are E -valued as stated, by direct evaluation of the four resolved pairs (Proposition A.1 of [5]). On E , $\wedge_k$ and $\vee_k$ act as Boolean AND/OR with $\text{CON} = \text{true}$, $\text{UNK} = \text{false}$. The first branch is active (its guard is CON) exactly when $x_2 = x_1$ or $x_3 = x_1$, in which case the majority value is x_1 ; the second branch is active exactly when both $x_2 \neq x_1$ and $x_3 \neq x_1$, in which case the majority value is Px_1 . The guard values absorb correctly on R : $\text{UNK} \wedge_k r = \text{UNK}$, $\text{CON} \wedge_k r = r$, $r \vee_k \text{UNK} = r$, so the inactive branch contributes UNK and the active branch passes its routed value through. Exhaustive evaluation over R^3 (machine-checked; Section 7) confirms the restriction is majority. $\square$

Remark 4.2 (the corrected count: routing data versus functions). A natural first count of the crown reads each coordinate as carrying an independent forward/backward routing choice and arrives at 2^n . That figure counts *routing data*, not functions: already at $n = 3$ the crown contains 16 functions (Theorem 3.3) — including ternary majority (Proposition 4.1), which is not of the form $\vec{x} \mapsto x_i$ or $\vec{x} \mapsto Px_i$ for any fixed i — strictly more than $2^3 = 8$. An earlier internal manuscript in the author's series stated the 2^n figure for this fragment; the present note supersedes it. The stable statement is Theorem 3.1: the crown is self-dual completeness, and its size is the size of the self-dual class.

5 One-Constant Completion

Theorem 5.1 (one resolved constant completes the outer fragment). *Fix either resolved constant, say $c = \text{FAL}$. For every map $g : R^n \rightarrow R$ (not necessarily equivariant) there exist a P -equivariant map $\hat{g} : R^{n+1} \rightarrow R$ with*

$$g(\vec{x}) = \hat{g}(c, \vec{x}),$$

where $\hat{g} \in G|_R^{(n+1)}$ by Theorem 3.1. Hence the clone generated on the face by $G|_R$ together with the constant c contains all 2^{2^n} maps $R^n \rightarrow R$.

Proof. In the Boolean reading ($c = 0$), define the self-dualization of g with one guard variable:

$$\hat{g}(x_0, \vec{x}) = \begin{cases} g(\vec{x}), & x_0 = 0, \\ \neg g(\neg \vec{x}), & x_0 = 1. \end{cases}$$

$\hat{g}$ is self-dual: complementing all $n + 1$ arguments swaps the two branches and complements the output —

$$\hat{g}(\neg x_0, \neg \vec{x}) = \begin{cases} g(\neg \vec{x}), & x_0 = 1, \\ \neg g(\vec{x}), & x_0 = 0, \end{cases}$$

which in both cases equals $\neg \hat{g}(x_0, \vec{x})$. Specializing the guard to the constant 0 gives $\hat{g}(0, \vec{x}) = g(\vec{x})$. By Theorem 3.1, $\hat{g}$ is the restriction of a closed-core term t of arity $n + 1$; substituting the constant c for the guard slot of t (a composition in the clone generated by $G|_R \cup \{c\}$) realizes g on R^n . The count 2^{2^n} is the number of all maps $R^n \rightarrow R$. $\square$

Remark 5.2. Theorem 5.1 is the zero/one accessibility contrast used by [5, App. A.5]: without constants the outer fragment is the proper self-dual class; a single resolved constant jumps it to everything. (In clone-theoretic terms this reflects the maximality of the self-dual clone in Post’s lattice [3]; the proof above is self-contained and does not invoke the classification.)

6 Sharpness: the Crown Is a Single-Orbit Phenomenon

Remark 6.1 (failure for larger resolved faces). Suppose the resolved face has $2k \geq 4$ elements, $R = O_1 \sqcup \cdots \sqcup O_k$ with P -pairs $O_i = \{a_i, b_i\}$. A resolved value now carries two pieces of information: the within-pair bit and the orbit label. P -equivariance constrains only the former. Closed-core terms, however, are orbit-confined: a unary term evaluated at a resolved r can output only values in $\{r, Pr\}$ together with the endpoints, because the generators $\wedge_k, \vee_k, P$ cannot manufacture an absolute orbit label from a single input. Concretely, on the six-element flat carrier with $R = \{a, b, c, d\}$, $P(a) = b$, $P(c) = d$, the unary map

$$h(a) = a, \quad h(b) = b, \quad h(c) = a, \quad h(d) = b$$

(extended by fixing the endpoints) is monotone, endpoint-preserving, and P -equivariant, yet no closed-core term restricts to it: unary closed-core terms act in one of the four global modes x , Px , $x \wedge_k Px$, $x \vee_k Px$, and none maps c into $\{a, b\}$. Hence for $|R| \geq 4$, P -equivariance strictly over-approximates the outer fragment, and the correct description is a routing/signature theory rather than the crown. The crown theorem is therefore stated here only for the dual-rail carrier, where R is a single P -orbit and the two notions coincide.

7 Mechanization

The note ships with a Lean 4 development that extends the published artifact of [5] (core Lean 4 only; no mathlib; no `native_decide`; no `sorry`; toolchain pinned; kernel axiom profile at most `[propext, Quot.sound]`):

- **necessity** — `crown_necessity`, from the mechanized Theorem 2.1 of [5] (`cterm_equivariant`);
- **sufficiency** — `crown_sufficiency`: the orbit-literal minterm DNF of Theorem 3.1, with the resolved/endpoint evaluation table machine-checked;
- **the count** — the explicit bijection (`restrictRep/extendSD` with both inverse laws) between self-dual maps of arity $n+1$ and free assignments on the first-coordinate-TRU representatives;
- **calibration** — `maj3_realized` (the explicit routing term computes ternary majority; kernel `decide` over the 8-point face) and `and2_not_selfDual`;
- **one-constant completion** — `sdGuard` with `sdGuard_selfDual`, `sdGuard_spec`, and the packaged `one_constant_completion`.

An independent replay script exhaustively checks the finite claims (detector tables, the majority term over R^3 , brute-force crown counts for $n \leq 4$, the 2^n miscount check, completion at $n \leq 2$, and the $|R| = 6$ counterexample of Section 6). The frozen output ships alongside.

References

- [1] Yu. I. Janov, A. A. Muchnik. *On the existence of k -valued closed classes without a finite basis*. Doklady Akademii Nauk SSSR, 127:44–46, 1959.
- [2] D. Lau. *Function Algebras on Finite Sets*. Springer, 2006. doi:10.1007/3-540-36023-9.
- [3] E. L. Post. *The Two-Valued Iterative Systems of Mathematical Logic*. Princeton University Press, 1941. doi:10.1515/9781400882366.
- [4] I. G. Rosenberg. *Complete sets of operations for finite algebras*. Mathematische Nachrichten, 44(1–6):253–258, 1970. doi:10.1002/mana.19700440120.
- [5] W. C. Yang. *The Price of NOT on D4*. Public edition v1.0, 2026. DOI: 10.5281/zenodo.21800033. <https://github.com/ycmath/price-of-not-on-d4>
- [6] W. C. Yang. *Finite-Energy Epistemic Logic with Conservative Pointed Extension and Negation Geometry*. Public edition v1.0, 2026. DOI: 10.5281/zenodo.21800031.

Authorship & provenance

Won Chul Yang, independent researcher. The mathematics of this note is the author’s, developed inside the author’s dual-rail carrier programme around Papers I–II [6, 5]; AI assistance (Anthropic Claude family) was used for the Lean 4 mechanization, the machine replay, and the preparation of the manuscript, with the Lean 4 kernel as the acceptance gate for the verification

layer. Remark [4.2](#) records a self-correction of the author's series: an earlier internal manuscript stated the 2^n count; the corrected function count $2^{2^{n-1}}$ is proved here. Corrections are invited.

Licenses: Apache-2.0 (Lean artifacts and verification scripts), CC BY 4.0 (text).